

THE TAXONOMIC STATUS OF *PANULIRUS ECHINATUS* SMITH, 1869
(DECAPODA MACRURA, PALINURIDAE)

BY

L. B. HOLTHUIS

Rijksmuseum van Natuurlijke Historie, Leiden

During the examination of the Palinuridae in the collection of the U. S. National Museum, Washington, D.C., two specimens of *Panulirus echinatus* Smith were found, the study of which showed that this species, which by many authors is considered to be of doubtful standing, is distinct from all other species of the genus.

***Panulirus echinatus* Smith, 1869**

Panulirus echinatus Smith, 1869: 20; Pocock, 1890: 516; Ortmann, 1891: 31; Rathbun, 1900: 151;

De Man, 1916: 33, 44; Holthuis, 1946: 110.

Palinurus echinatus Von Martens, 1872: 125; Pfeffer, 1881: 31.

Panulirus échinatus Gruvel, 1911: 29.

Panulirus guttatus brasiliensis Faria & Silva, 1937: 44, phot. 2, 6, 7, 8, figs. 12-18.

Material examined:

Recife (= Pernambuco), Brazil; 1 July 1899; A. W. Greeley coll.; Branner-Agassiz Expedition.
— 1 ♂, cl. 53 mm.

Recife (= Pernambuco), Brazil; 1875-1877; C. F. Hartt Exploration. — 1 ♂, cl. 49 mm.

The status of *Panulirus echinatus* Smith has been the object of much difference of opinion. In the original publication Smith gave an extensive description of the species, but did not provide any illustrations of it. He pointed to the close resemblance of his new species to *Panulirus guttatus* (Latreille) of which, unfortunately, he had no actual material for comparison and therefore had to rely upon H. Milne Edwards's (1837: 297, pl. 23 figs. 1, 2) account of the species. Smith already mentioned as distinguishing character between the two species the fact that the grooves of the abdominal somites are interrupted in the middle in *P. echinatus*, while they are entire in *P. guttatus*. As already pointed out by Pfeffer (1881: 31), the other differences mentioned by Smith are either unreliable or based on inaccuracies in Milne Edwards's figures of *P. guttatus*. Pfeffer (1881: 32) did not attach much value to the character of the uninterrupted grooves and came to the conclusion "dass der *P. echinatus* nicht als Art, vielleicht jedoch als Lokalforn von *P. guttatus* anzusehen ist". Ortmann (1891: 30, 31) placed *P. echinatus* definitely in the synonymy of *P. guttatus*. That Pfeffer and Ortmann attached so

little value to the interruption of the abdominal grooves is mainly due to the fact that they confused various species under the name *P. guttatus* and for that reason ascribed a wider range of variation to the various characters. Neither Pfeffer nor Ortmann had seen material of *P. echinatus*. Pocock (1891) and Rathbun (1900), who did examine material of that species, came to the conclusion that it is a valid species distinct from *P. guttatus*. Gruvel (1911) in his monographic treatment of the Palinuridae synonymized the two species again, while De Man (1916) and Holthuis (1946), who gave a list of the Palinuridae known to them, both treat *P. echinatus* as a distinct, though rather dubious species.

In a rather obscure publication, which has escaped the notice of most carcinologists¹⁾, Faria & Silva (1937) gave a very important account of the Brazilian Palinuridae. They recognized *P. echinatus* as distinct from *P. guttatus*, but believed the differences not sufficiently important to allow the two forms to stand as separate species. They considered Smith's species a variety of *P. guttatus* for which they proposed, quite unnecessarily, the new name *Panulirus guttatus brasiliensis*. They gave a good description and several figures of this variety.

An examination of the two specimens of *P. echinatus* in the collection of the U. S. National Museum, one of which is the specimen dealt with by Rathbun (1900), convinced me that Smith, Pocock and Rathbun were right in considering *P. echinatus* a full species distinct from *P. guttatus*. *Panulirus echinatus* is most closely related to *P. guttatus* and *P. penicillatus*. With these two species it agrees in the development of the exopods of the maxillipeds: in the first and second maxillipeds the exopod is well developed with a long flagellum, while in the third maxilliped the exopod is present but bears no flagellum at all. In all other species of the genus the exopod of the third maxilliped is either entirely absent (or at most reduced to a minute bud), or it is provided with a well developed flagellum. *Panulirus penicillatus* (Olivier), a species from the Indo-West Pacific and East Pacific regions, is immediately distinguished from the other two species in that the antennular somite bears two pairs of spines dorsally; these pairs are placed close together at their bases but diverge at the top. In *P. guttatus* and *P. echinatus*, both of which are restricted to the eastern Atlantic, the antennular somite bears only one pair of spines.

A comparison of the above specimens of *Panulirus echinatus* with actual specimens of *P. guttatus* (9 miles S. W. of Key West, Florida, in lobster trap, 8 fathoms, 19 December 1946, G. Gunter & C. E. Dawson. — 1 ♂; Havana Bay, Cuba, M. Sanchez Roig. — 1 ovig. ♀; San Geronimo, San Juan, Puerto Rico, reef and tidepools, flashlight at night, 30 April 1937, Smithsonian-Hartford Expedition, Sta. 74.— 2 ♂♂, 1 ovig. ♀; Caledonia Bay, Panama, 8°53'38" N 77°41'25" W, rocky reef, shore collecting, 26 April 1959, Allan Hancock Exped.

¹⁾ The only reference in the literature to this publication, which I so far have found, is that by P. Sawaya (1942, *Marcgrave's Hist. nat. Bras.*: lxiii, lxiv). Professor Sawaya was so kind as to provide me with a photocopy of Faria & Silva's paper, for which I wish to express here my sincere thanks.

Sta. A. 50-39.— 1 ♂; Bath Reef, off Bayview, Barbados, 22 July 1959, A. G. Humes & R. U. Gooding. — 1 ovig. ♀) showed the following differences between the two species:

1. The spinulation of the carapace is very similar, but in *P. echinatus* there are more small spinules in the anterior part of the carapace in front of the cervical groove.

2. The transverse groove on abdominal somites II to V inclusive, is interrupted in *P. echinatus*; it is always continuous in *P. guttatus*. Smith described only the third to fifth somites of *P. echinatus* as having an interrupted groove, while Faria & Silva found that the groove of the second somite is interrupted in the females, entire in the males. In my two male specimens the groove of the second somite is interrupted, although the gap is smaller than in the following somites. Laterally

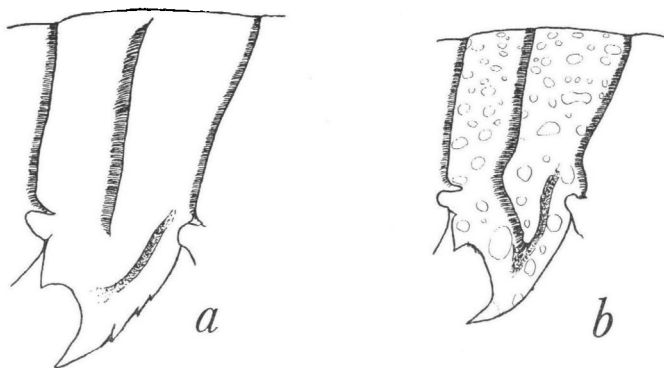


Fig. 1. Lateral view of second abdominal somite. a, *Panulirus echinatus* Smith from Recife (Branner-Agassiz Expedition). b, *Panulirus guttatus* (Latreille) from 9 miles S.W. of Key West. a, $\times 2$; b, $\times 1.3$.

the transverse groove of the second to fifth abdominal somites of *P. echinatus* does not reach the anterior groove of the pleura; in *P. guttatus* these two grooves join each other (fig. 1).

3. The anterior margin of the first to fifth pleura of the abdomen of *P. echinatus* bears one or more distinct sharply pointed teeth, while in *P. guttatus* there are at most a few tubercles there which seldom are pointed.

4. The antennulae and antennae are relatively shorter in *P. echinatus*. The first antennular segment is definitely shorter than the distance between the cervical groove and the anterior margin of the carapace, measured in the median line. In *P. guttatus* the first antennular segment is equal to or far longer than the distance between the cervical groove and the anterior margin.

5. The spinulation of the antennal peduncle is practically the same in the two species, except that the outer surface of the last segment in *P. echinatus* is smooth, while in *P. guttatus* it bears 2 to 4 spines placed in a longitudinal row.

6. The legs are much shorter in *P. echinatus* than in *P. guttatus*.

7. The sternum in *P. echinatus* is flatter and less deeply sculptured than that

of *P. guttatus*. In the former species its posterior margin bears two short blunt tubercles medially instead of the two rather slender diverging teeth shown in *P. guttatus*.

8. Rathbun (1900: 151) described the colour of her specimen of *P. echinatus* as follows: "Carapace, antennae, first two segments of abdomen and upper half of meral joints of legs pinkish red; remainder of abdomen and legs dark blackish blue. No round spots as in *P. guttatus* (Latreille)". This colour, part of which (the bluish legs) is still to be observed in the specimen, evidently is not the natural colour, which is described by Faria & Silva (1937: 45) as follows: "violeta claro na porção dorsal e amarelo claro na porção ventral; pedunculos das antenas com porções de côr violeta e outras de côr amarela; pedunculos das antenulas com faixas longitudinaes das côres amarelo e violeta; pernas tambem com faixas longitudinaes das mesmas côres que as das antenulas nos quatro primeiros segmentos; espinhos post-oculares transfaciados de amarelo e violeta; abdomen semeado profusamente de manchas arredondadas, de tamanhos variados de côr amarelo claro." In having the abdomen provided with many light roundish spots, *P. echinatus* thus resembles *P. guttatus*. The main colour difference between the two species seems to be that in *P. guttatus* the merus of the pereopods is spotted, while in *P. echinatus* it is streaked with yellow and purple.

So far *P. echinatus* has only been reported from the Brazilian coast. The records in the literature are as follows: Recife (= Pernambuco), N. E. Brazil (Smith, 1869; Rathbun, 1900), Fernando Noronha (Pocock, 1890), Atoll das Rocas, W. of Fernando Noronha, 3°50' S 33°50' W (Faria & Silva, 1937).

Faria & Silva (1937) gave an interesting account of the biology of *Panulirus echinatus* from which the following data are taken. The species lives in natural holes in the coral reefs surrounding the islands Pharol and Cimeterio of the Atoll das Rocas. The animals were observed to leave the shelter of their holes at high tide when they went out in search of food. Of the 856 lobsters examined in the period from 1 July to 7 August 1935, 138 were males and 718 females; 217 of the latter proved to be ovigerous. Between 4 and 6 August 1935, 394 specimens were examined, 76 of these being ovigerous with eggs in different stages of development. The females are as a rule smaller than the males: the majority of the specimens of 125 to 250 mm long proved to be females, while those between 255 and 360 mm were mostly males. The largest specimen was 360 mm long; the smallest ovigerous female measured 170 mm.

ZUSAMMENFASSUNG

Panulirus echinatus Smith, 1869, ist eine gute Art, die sich von *P. guttatus* (Latreille) in einer Anzahl konstanter Merkmale unterscheidet. Die Smith'sche Art wurde von Faria & Silva (1937) ausführlich beschrieben und abgebildet unter dem Namen *Panulirus guttatus brasiliensis*.

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